

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Define Problem Statements

ShopEZ - MERN E-Commerce Application

Phase: Brainstorming & Ideation Phase

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Classification: Project Documentation — For Academic Submission

Define Problem Statements

1. Introduction

This document presents the problem statements identified during the Ideation Phase for the ShopEZ E-Commerce Application.

The project aims to provide users with a centralized online shopping platform where they can browse products, register accounts, securely log in, manage wishlists, add products to carts, and complete purchases efficiently.

The problem statements are structured using the "I am / I'm trying to / But / Because / Which makes me feel" framework to understand user needs and challenges in existing online shopping platforms.

2. Problem Statements

PS-1: Online Shopper

Field	Description
I am	A customer who wants to purchase products online.
I'm trying to	Find products easily and complete purchases quickly
But	Many websites have complicated navigation and poor user experience
Because	Products are not organized properly and search functionality is limited.
Which makes me feel	Frustrated and less interested in completing purchases.

Impact Severity: ● High - Poor user experience can directly affect sales and customer satisfaction.

PS-2:Registered User

Field	Description
I am	A registered user of an e-commerce platform
I'm trying To	Manage my profile, cart, and wishlist efficiently
But	Existing systems do not provide seamless management features.
Because	User data and shopping activities are scattered across multiple pages
Impact Severity: 2 Medium to High –Poor account management reduces customer engagement.	

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3. Root-Cause Analysis (Ishikawa Diagram)



4. Problem-to-Feature Mapping

Root Cause	Proposed Solution Feature
Difficult product browsing	Product Listing Page
Lack of user authentication	Secure Registration & Login
Poor cart management	Shopping Cart Feature
No wishlist support	Wishlist Module
Product update difficulties	CRUD Operations
Lack of centralized data	MongoDB Database
Weak security	JWT Authentication
Poor checkout experience	Checkout Page

5. Conclusion

The ShopEZ E-Commerce Application provides a complete online shopping solution by addressing challenges faced by users during product discovery, account management, and purchasing. The platform improves user experience through secure authentication, product management, cart functionality, wishlist support, and CRUD operations while ensuring efficient data management using MongoDB.

